



SMART WASTE MANAGEMENT SYSTEM

Team No: 10 | Section: S3 | Members: Ili. Harshika – 2520030607, Angela Jose – 2520030365, Chinta Pavan Kumar – 2520030312 | Guide: Dr. Spandana S, Assistant Professor, Dept. of CSE

Database Systems Engineering and Distributed Backend Development (25CS1302E) | Y25 – 2026-2027 | Koneru Lakshmaiah Education Foundation (KLH University), Bachupally Campus, CSE

Abstract

- A database-driven system that uses simulated real-time waste-level data to monitor smart bins and identify bins crossing predefined thresholds.
- It manages collection requests from creation to completion and keeps historical records for reports and better collection planning.

Problem Statement & Objectives

Problem:

- Fixed schedules and manual monitoring cause overflowing bins and unnecessary trips.
- Fuel and manpower are wasted, and collection records are not organized.

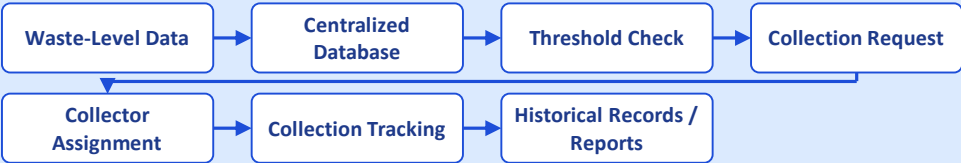
Objectives:

- Monitor waste levels using simulated data.
- Identify bins crossing thresholds.
- Create, assign and track collection requests.
- Maintain records and generate reports.

Existing vs Proposed System

Feature	Existing System	Proposed System
Monitoring	Manual monitoring	Waste-level monitoring
Collection	Fixed schedules	Threshold-based collection
Records	Limited records	Centralized records
Planning	Possible unnecessary trips	Request tracking

Proposed Workflow



System Modules

1 User & Role Management
Users, roles and access

2 Smart Bin Management
Bin location, capacity and status

3 Waste-Level Monitoring
Simulated readings and threshold detection

4 Collection Management
Requests, assignment and tracking

5 Collector & Vehicle Management
Collector details, vehicles and assignments

6 Reports & Analytics
Statistics and historical reports

Tools & Technologies

DATABASE
PostgreSQL

BACKEND
Node.js + Express.js

FRONTEND
React.js

MAPS
Leaflet + OpenStreetMap

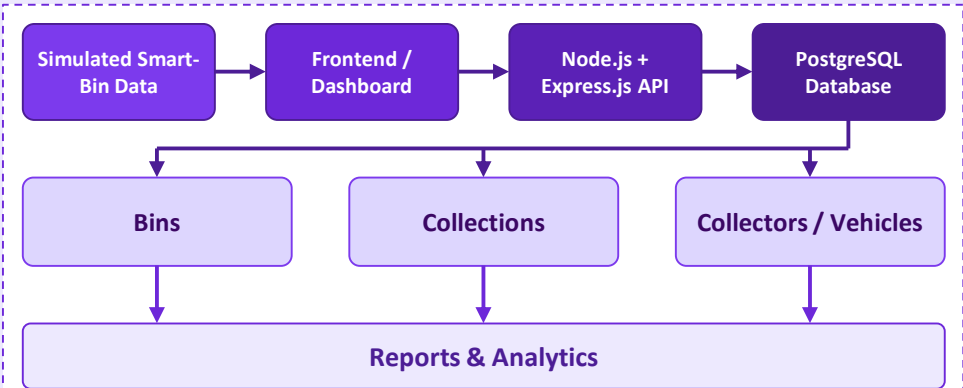
API TESTING
Postman

DEVELOPMENT
Visual Studio Code

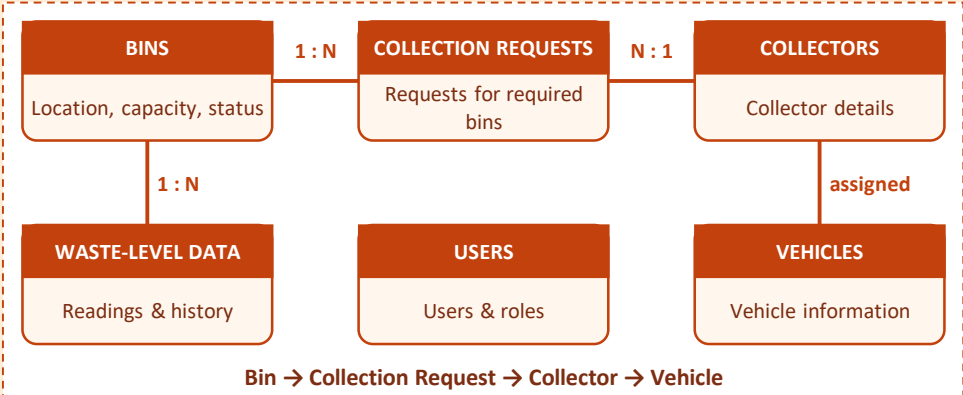
CONTAINERIZATION
Docker Desktop

DATA SIMULATION
Node.js

System Architecture



ER Diagram & Database Design



Implementation Screenshots

Dashboard Overview

Route Optimization

Results & Testing

- Centralized waste-bin information
- Threshold-based bin identification
- Collection request tracking
- Collector assignment
- Historical data management
- Report generation

Conclusion

- A structured, data-driven approach to waste collection.
- Monitors bin conditions, identifies collection needs and maintains organized collection records.

Future Scope

- Physical IoT sensors and GPS vehicle tracking
- Advanced route optimization
- Mobile app and critical-bin notifications
- Predictive waste-generation analysis
- Cloud deployment and smart-city integration